

SOAL PRE-BOOTCAMP WB CLASS

Data Types:

Problem 1: Even or Odd Sum

For Example You're given a list of

[2, 57, 18, 60, 17, 1, 6, 59, 19, 64, 87, 17, 69, 83, 92, 55, 5, 12, 22, 73, 25, 83]

Compute the sum of all even number in the list!

Problem 2: String Length Counter

Create a Python program that takes a list of strings as input and returns the length of each string as a new list.

Problem 3: Finding Unique Elements

Write a Python function that takes a list of numbers as input and returns a new list containing only the unique elements (no duplicates).

Looping:

Problem 1: Factorial Calculator

Write a Python function that takes a non-negative integer as input and calculates its factorial using a loop.

Problem 2: Pattern Printer

Create a Python program that takes an integer as input and prints a pattern of asterisks in the form of a right-angled triangle. The user specifies the number of rows.

Problem 3: Palindrome Detector

Write a Python function that takes a string as input and checks if it is a palindrome (reads the same forwards and backwards).

Functions:

Problem 1: Circle Area Calculator

Create a Python function that calculates and returns the area of a circle when given its radius as input.

Problem 2: String Reverser

Write a Python function that takes a string as input and returns the reverse of the string.

Problem 3: Exponent Calculator

Create a Python function that takes two numbers as input and computes the first number raised to the power of the second number.

Indexing and Slicing:

Problem 1: Middle Element(s)

Write a Python function that takes a list as input and returns the middle element (or elements if the list has an even number of elements).

Problem 2: Vowel Extractor

Create a Python program that takes a string as input and extracts and prints all the vowels (a, e, i, o, u) from the string.

Problem 3: List Reverser

Write a Python function that takes a list as input and returns a new list with the elements reversed.

Basic Operation:

Problem 1: Calculator Function

Write a Python function that can perform basic arithmetic operations like addition, subtraction, multiplication, and division. It should take two numbers and an operator as input and return the result. Ensure proper handling of division by zero.

Problem 2: Simple Interest Calculator

Create a Python program that calculates and prints the simple interest for a given principal amount, rate of interest, and time period.

Problem 3: Fibonacci Series

Write a Python function that generates and returns the first N numbers in the Fibonacci series, where N is the input from the user. The Fibonacci series starts with 0 and 1, and each subsequent number is the sum of the previous two.

Vectors:

Problem 1: Vector Addition

Write a Python function that takes two vectors as input (represented as lists of numbers) and returns their sum as a new vector.

Problem 2: Vector Dot Product

Create a Python function that calculates and returns the dot product of two vectors.

Problem 3: Vector Magnitude

Write a Python function that calculates and prints the magnitude (length) of a vector represented as a list of numbers.

Matrices:

Problem 1: Matrix Addition

Create a Python function that takes two matrices as input (represented as lists of lists) and returns their sum as a new matrix.

Problem 2: Matrix Transposition

Write a Python program that transposes a given matrix (swaps rows with columns) and displays the transposed matrix.

Problem 3: Matrix Multiplication

Create a Python function that multiplies two matrices and returns the result as a new matrix. Ensure that the input matrices are compatible for multiplication.

FINAL

Problem: Analyzing Student Performance Data

You are provided with a dataset containing information about students' academic performance. The dataset includes the following columns: Student ID, Age, Gender, Math Score, Reading Score, Writing Score, and Lunch Type (Standard or Free/Reduced).

Your task is to analyze this dataset to gain insights into student performance, relationships between variables, and create various statistical plots to visualize the data.

Part 1: Data Analysis

Calculate and report the following central tendencies for Math, Reading, and Writing Scores:

- Mean
- Median
- Mode

Calculate the following measures of dispersion for each of the three scores (Math, Reading, and Writing):

- Range
- Variance
- Standard Deviation
- Interquartile Range (IQR)

Describe the data:

- Calculate the minimum and maximum ages of the students.
- Determine the gender distribution among students.
- Identify the most common lunch type among students.
- Explore the relationship between Math, Reading, and Writing Scores by calculating the Pearson correlation coefficients between pairs of scores. Comment on the correlations found.

Part 2: Data Visualization

Create a histogram for the Math Scores. Label the axes and title the plot appropriately.

Generate a line chart that shows how student scores (Math, Reading, and Writing) change with age. The x-axis should be the students' ages, and the y-axis should be the respective scores. Include a legend to distinguish between the three scores.

Design a bar chart to compare the average Math Scores for students with Standard and Free/Reduced lunch types. Label the axes and provide a title.

Construct a scatterplot that visualizes the relationship between Math and Reading Scores. Use different colors to distinguish between students with Standard and Free/Reduced lunch types. Add labels and a legend.

Finally, create a statistical plot (e.g., box plots or violin plots) for each of the three scores (Math, Reading, and Writing) to show their distributions and any potential outliers.

Part 3: Interpretation

Provide a summary of your findings. Discuss any interesting insights you've gathered from the central tendency, dispersion, data description, correlation, and data visualization analysis.